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Choi et al.

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(54) **CAMERA MODULE INCLUDING FOLDED OPTICS**

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Primary Examiner — Wyatt A Stoffa

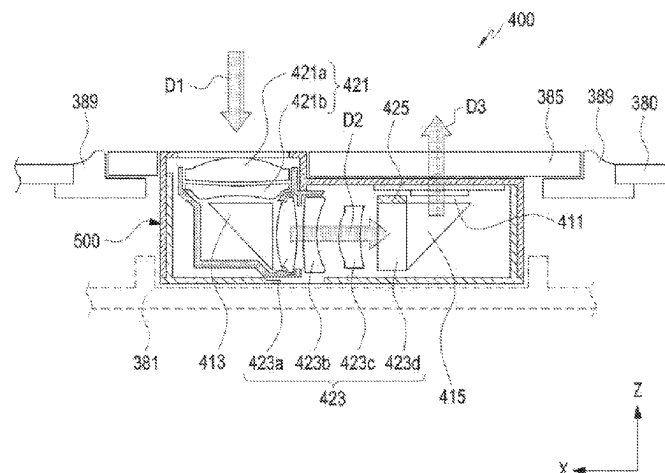
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(57) **ABSTRACT**

A camera module includes: a first refractive member configured to reflect or refract light that is incident on the first refractive member in a first direction, in a second direction intersecting the first direction; a second refractive member configured to reflect or refract the light that is reflected or refracted by the first refractive member and is incident on the second refractive member in the second direction, in a third direction intersecting the second direction; an image sensor configured to detect the light that is reflected or refracted by the second refractive member and is incident on the image sensor in the third direction; and at least one optical member provided between the first refractive member and the second refractive member, wherein the at least one optical member is configured to block the light that is reflected or refracted by the first refractive member from being incident on the image sensor, and to guide the light that is reflected or refracted by the first refractive member to be incident on the second refractive member in the second direction.

16 Claims, 9 Drawing Sheets



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See application file for complete search history.

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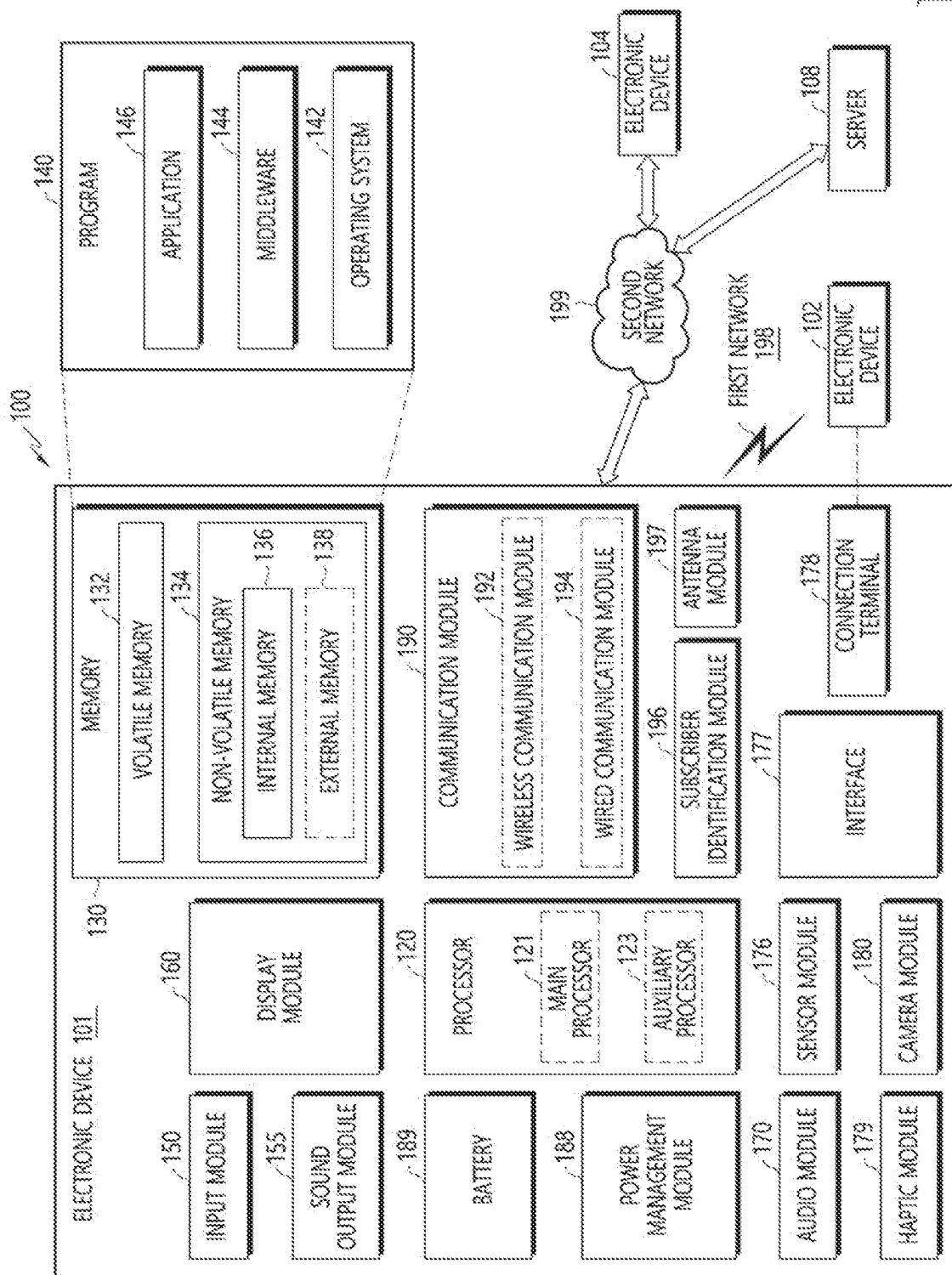


FIG. 1

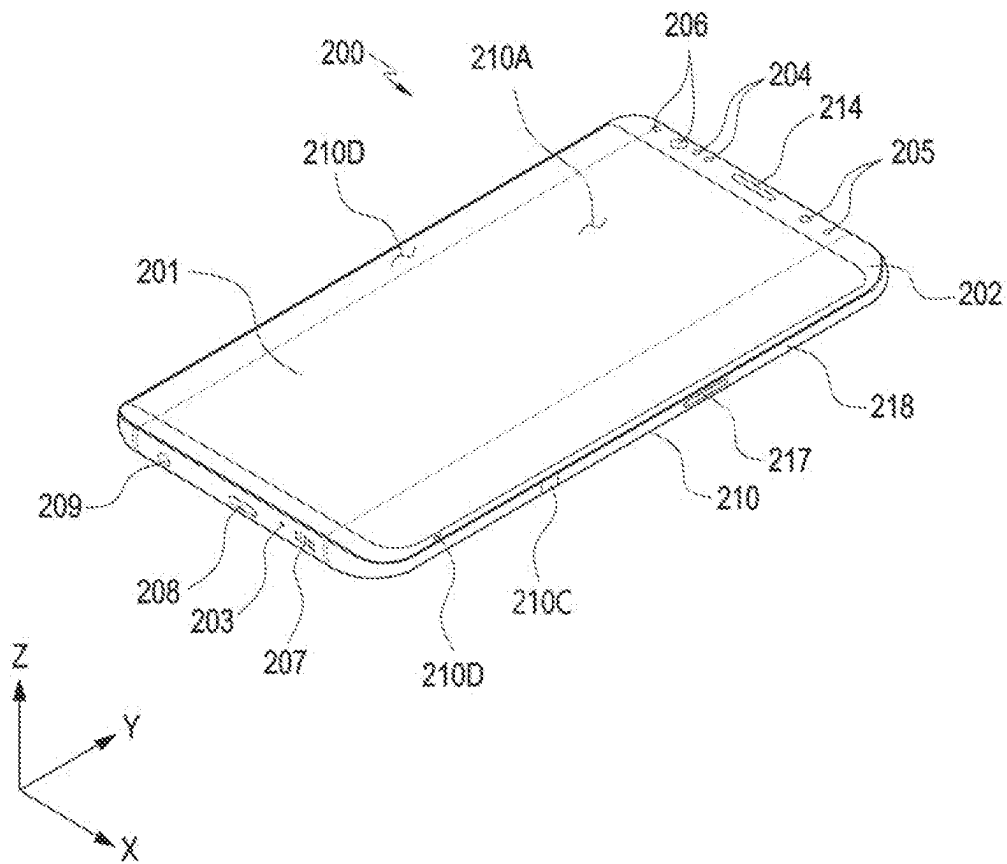


FIG. 2

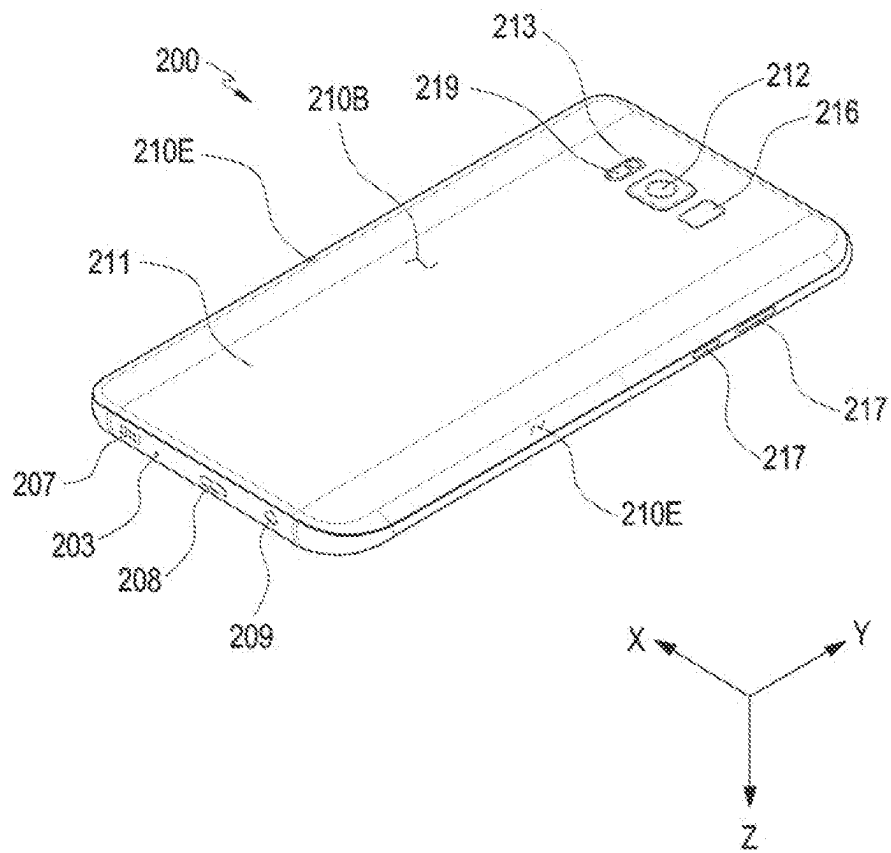


FIG. 3

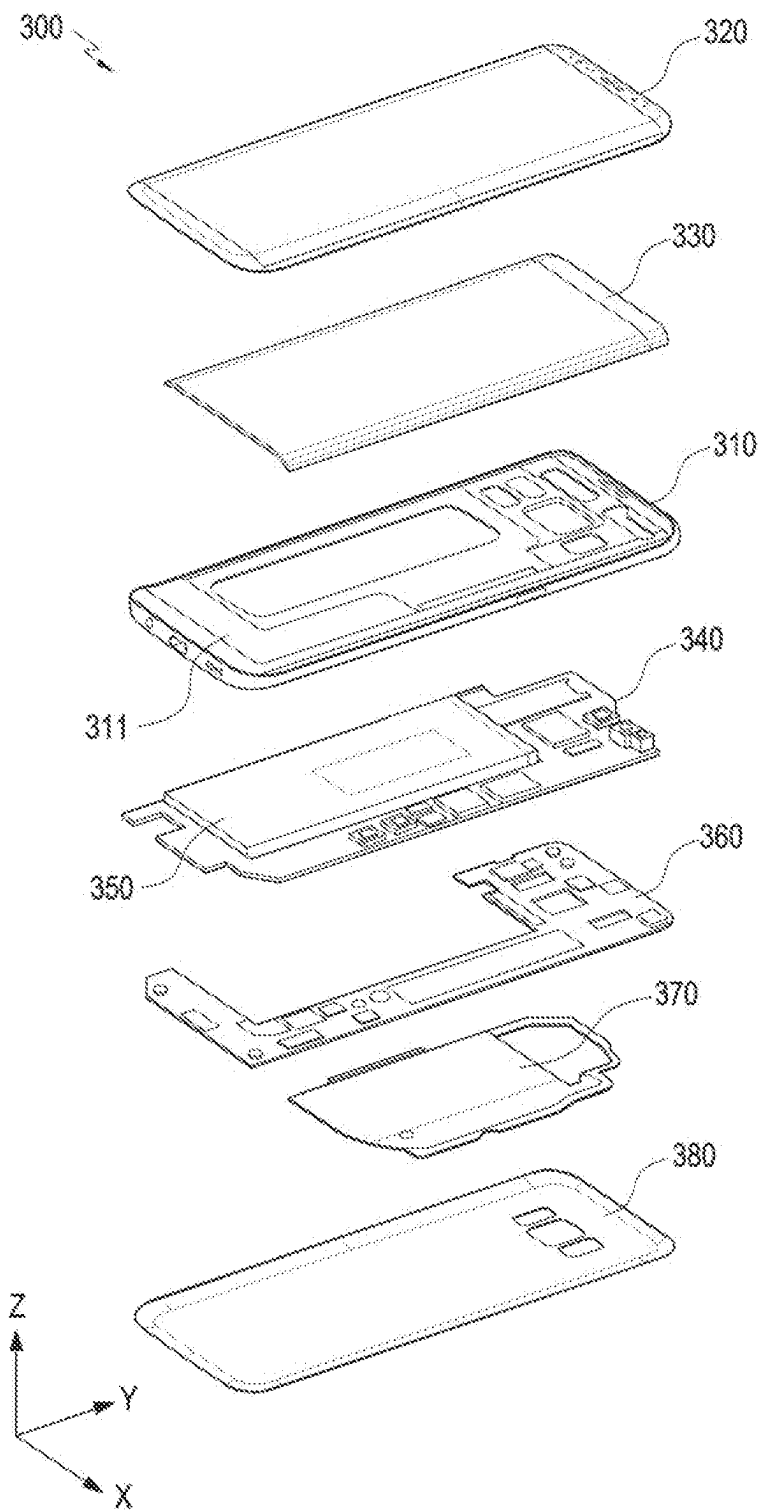


FIG.4

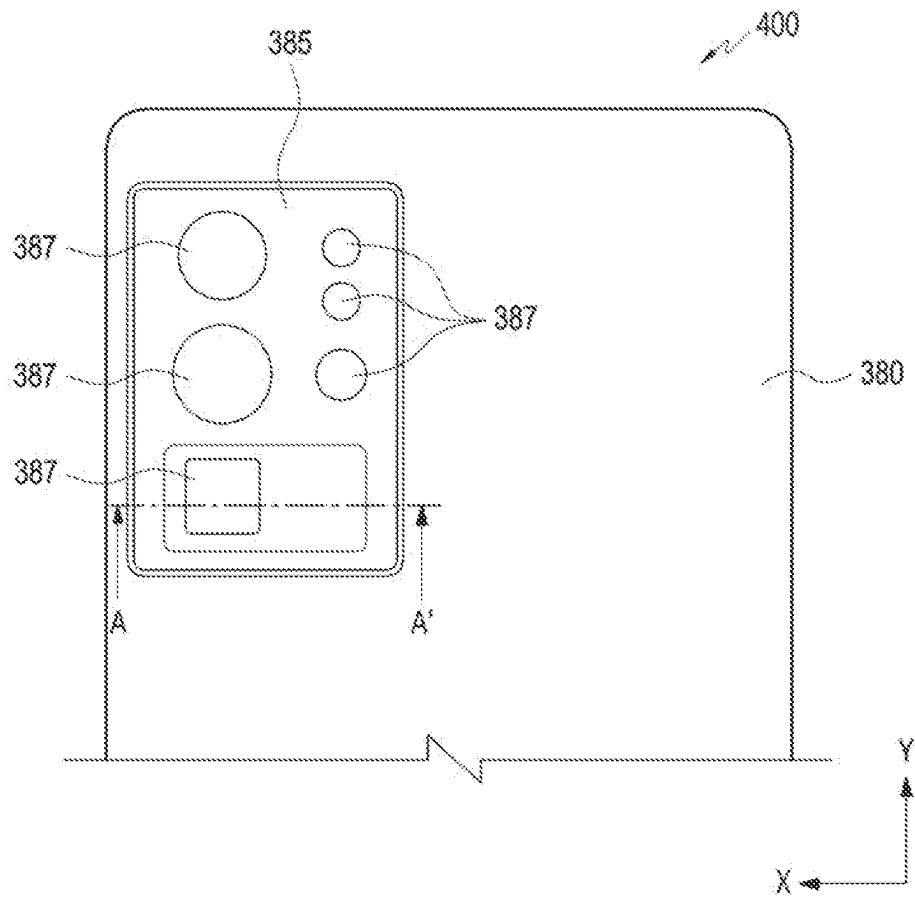


FIG.5

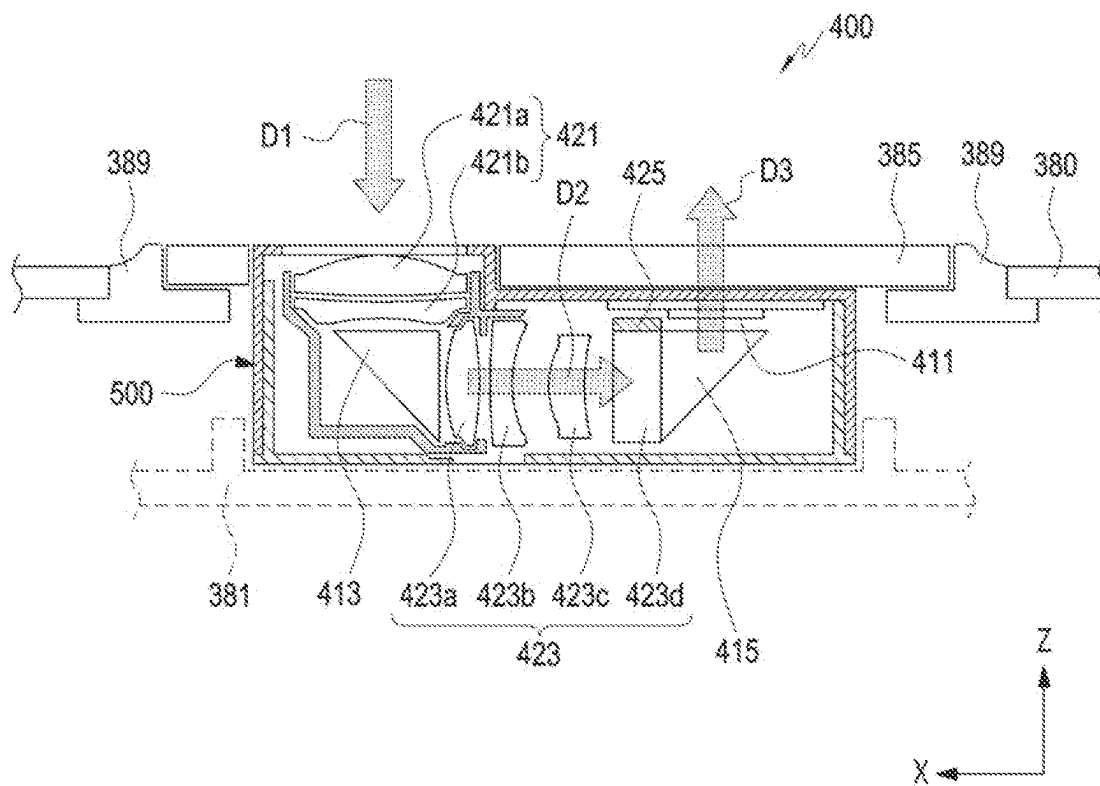


FIG. 6

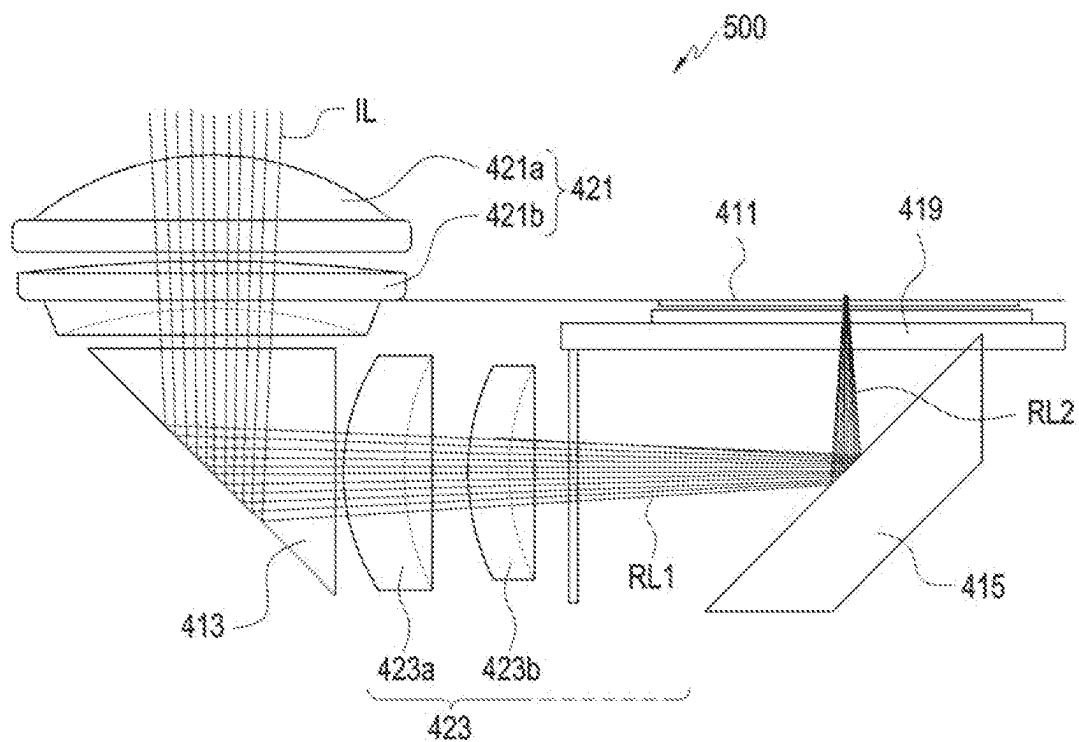


FIG. 7

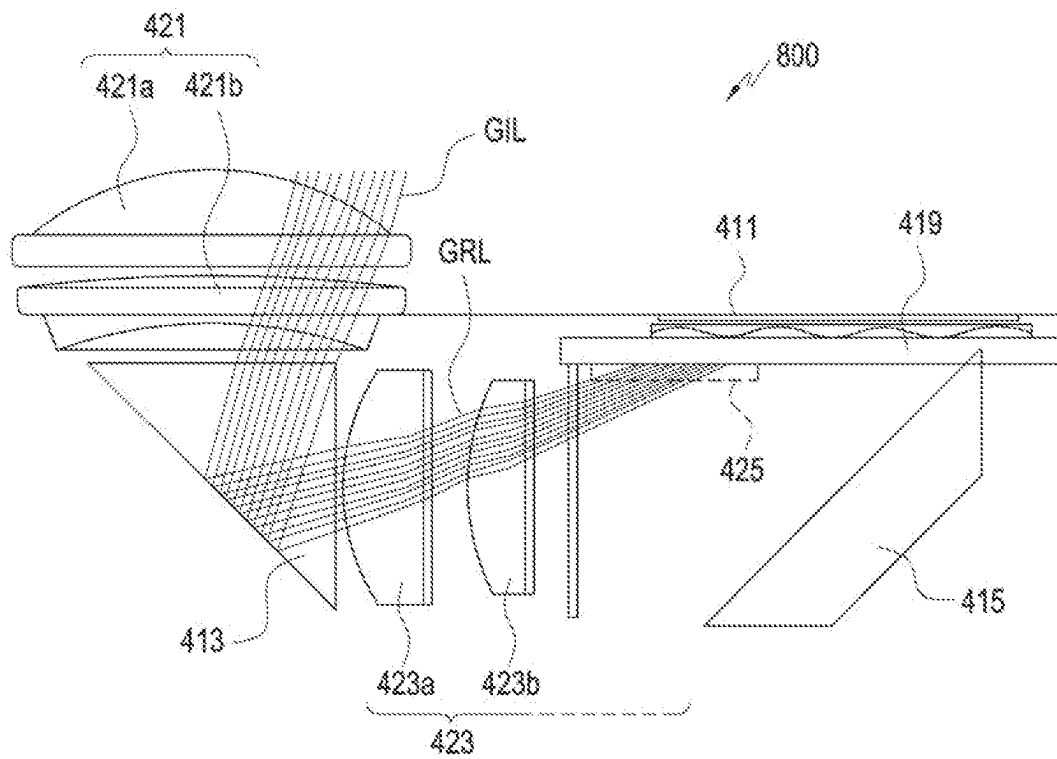


FIG. 8

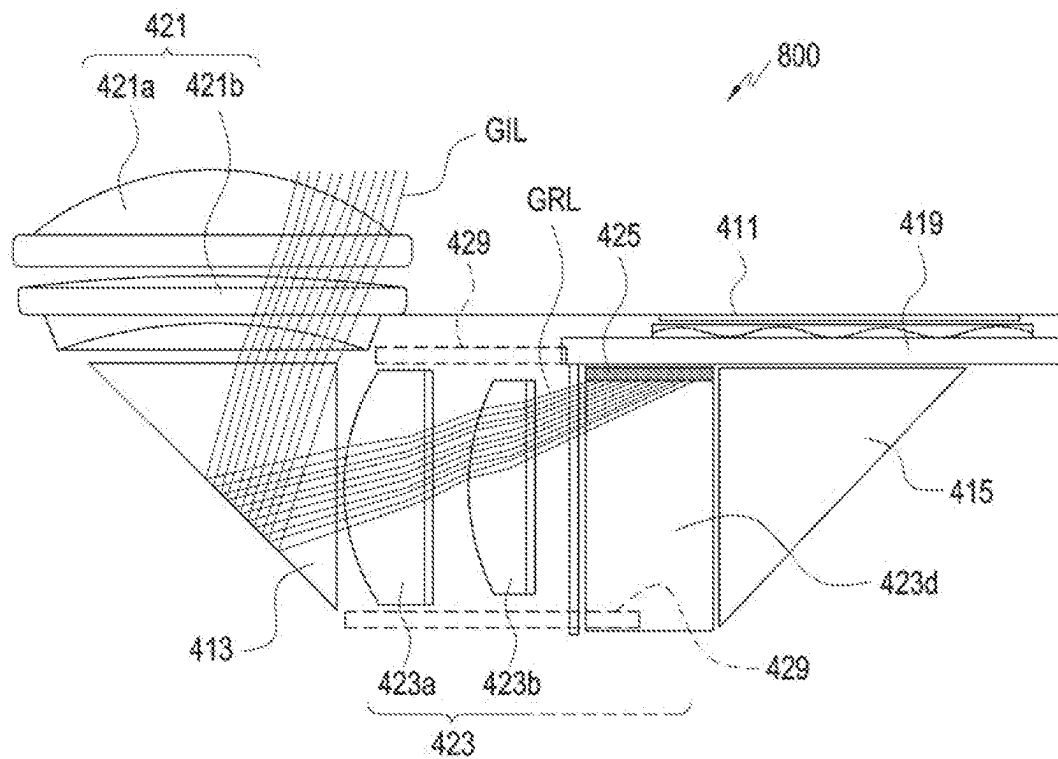


FIG. 9

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CAMERA MODULE INCLUDING FOLDED OPTICS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation application, claiming priority under § 365(c), of an International application No. PCT/KR2022/012004, filed on Aug. 11, 2022, which is based on and claims priority to Korean Patent Application No. 10-2021-0108053, filed on Aug. 17, 2021 in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND

Technical Field

The disclosure relates to a lens assembly or a camera module, for example, to a camera module including a folded optics.

Background Art

An electronic device may be a device that performs a specific function according to a program provided therein (e.g., an electronic scheduler, a portable multimedia reproducer, a mobile communication terminal, a tablet personal computer (PC), an image/sound device, a desktop/laptop PC, or a vehicle navigation system), as well as a home appliance. The above-mentioned electronic devices may output, for example, information stored therein as sound or an image. As the degree of integration of electronic devices has increased and super-high-speed and large-capacity wireless communication has become popular, multiple functions have recently come to be provided in a single electronic device, such as a mobile communication terminal. For example, various functions, such as an entertainment function such as a game function, a multimedia function such as a music/video play function, a communication and security function for mobile banking or the like, and/or a schedule management or e-wallet function, are integrated in a single electronic device, in addition to a communication function.

With the development of digital camera manufacturing technology, electronic devices equipped with small and lightweight camera modules have been commercialized. As an electronic device (e.g., a mobile communication terminal) that is generally carried at all times is equipped with a camera module, it becomes possible for a user to easily utilize various functions such as video call or augmented reality as well as to take a picture or video.

In recent years, electronic devices including a plurality of cameras have been distributed. An electronic device may include, for example, a camera module including a wide-angle camera and a telephoto camera. The electronic device may acquire a wide-angle image by photographing a wide-range scene around the electronic device using the wide-angle camera, or may acquire a telephoto image by photographing a scene corresponding to a location relatively far from the electronic device using the telephoto camera. In this way, by including a plurality of camera modules or lens assemblies, miniaturized electronic devices such as smart phones are making inroads into the compact camera market, and are expected to replace high-performance cameras such as single-lens reflex cameras in the future.

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SUMMARY

Technical Problem

5 In a miniaturized electronic device including a plurality of camera modules, a folded optics camera may be useful for extending or enlarging a focal length. For example, in a folded camera, since an optical member such as a prism is disposed, the direction in which lenses are arranged can be freely designed or arranged, regardless of the direction in which external light is incident, and thus it is easy to extend the focal length. As the degree of design freedom for the arrangement direction of lenses is improved in folded cameras, a miniaturized telephoto camera can be implemented, and can be mounted in an electronic device in combination with a wide-angle camera. In general, by including an optical image stabilizer in a camera, the quality of an acquired image can be improved, and the performance of a telephoto camera for photographing a distant subject can be greatly improved when the optical image stabilization function is combined therewith.

15 In a camera of a miniaturized and/or lightweight electronic device, by implementing the optical image stabilization function in a manner of shifting an image sensor, which is lighter and smaller in volume than optics, it is possible to secure good power efficiency while saving installation space. However, in a miniaturized electronic device, it may be difficult to secure an installation space for a camera in which a telephoto function and an optical image stabilization function are combined.

25 One or more embodiments of the disclosure provide at least solve the above-described problems and/or disadvantages and provide at least the advantages described below, and are able to provide a lens assembly and/or a camera module that facilitates implementation of a telephoto function in a miniaturized electronic device.

30 One or more embodiments of the disclosure provide a lens assembly and/or a camera module that facilitates implementation of an image stabilization function combined with a telephoto function in a miniaturized electronic device.

35 One or more embodiments of the disclosure provide a lens assembly and/or a camera module in which a high-performance image sensor is easily mounted while having a telephoto function and an image stabilization function in a miniaturized electronic device.

40 The problems to be solved in the disclosure are not limited to the above-mentioned problems, and may be variously expanded without departing from the spirit and scope of the disclosure.

45 The problems to be solved in the disclosure are not limited to the above-mentioned problems, and additional aspects according to various embodiments will be presented through the detailed description to be described later, and may be partially clear from the description or understood through the presented embodiments.

50 According to an aspect of the disclosure, there is provided a camera module including: a first refractive member configured to reflect or refract light that is incident on the first refractive member in a first direction, in a second direction intersecting the first direction; a second refractive member configured to reflect or refract the light that is reflected or refracted by the first refractive member and is incident on the second refractive member in the second direction, in a third direction intersecting the second direction; an image sensor configured to detect the light that is reflected or refracted by the second refractive member and is incident on the image sensor in the third direction; and at least one optical member

provided between the first refractive member and the second refractive member, wherein the at least one optical member is configured to block the light that is reflected or refracted by the first refractive member from being incident on the image sensor, and to guide the light that is reflected or refracted by the first refractive member to be incident on the second refractive member in the second direction.

The at least one optical member may include: a dummy member extending in the second direction and configured to transmit light traveling in the second direction; and a light shielding member provided on at least a portion of an outer circumferential surface of the dummy member, wherein the light shielding member is configured to absorb, scatter or reflect the light that is reflected or refracted by the first refractive member to block the light from being incident on the image sensor.

The camera module may further include a lens group having a positive refractive power, wherein the lens group is configured to guide or focus, to the first refractive member, the light that is incident in the first direction.

The at least one optical member may include at least one lens configured to move forward and backward between the first refractive member and the second refractive member along the second direction.

The at least one optical member may further include a light shielding member provided on at least a portion of an outer circumferential surface of the at least one lens, and configured to absorb, scatter, or reflect the light that is reflected or refracted by the first refractive member to block the light from being incident on the image sensor.

The camera module may further include an infrared cut-off filter provided between the second refractive member and the image sensor, and the image sensor may be configured to detect light that passes through the infrared cut-off filter.

At least one of the first refractive member, the second refractive member, or the at least one optical member may include an infrared cut-off coating layer, and the image sensor may be configured to detect light that passes through the infrared cut-off coating layer.

The second direction may be perpendicular to the first direction, and the third direction may be perpendicular to the second direction.

The image sensor may be configured to horizontally move in a plane perpendicular to the first direction or a plane perpendicular to the third direction.

An effective focal length may be 1.8 times or more and 2.5 times or less of a diagonal length of an imaging plane of the image sensor.

The at least one optical member may include a light shielding member configured to at least partially absorb, scatter, or reflect the light that is reflected or refracted by the first refractive member and travels toward the image sensor.

The camera module may further include at least one lens configured to move forward and backward along the second direction between the first refractive member and the second refractive member, wherein the at least one lens is configured to guide or focus, to the second refractive member, the light that is reflected or refracted by the first refractive member.

The camera module may further include an infrared cut-off filter provided between the second refractive member and the image sensor, wherein the light shielding member is provided on at least a portion of the infrared cut-off filter, and the image sensor is configured to detect light that passes through the infrared cut-off filter.

The camera module may further include: a lens group having a positive refractive power and configured to guide or focus, to the first refractive member, the light incident on the first refractive member in the first direction; and at least one lens provided between the first refractive member and the second refractive member, the at least one lens being configured to move forward and backward along the second direction, wherein the at least one lens is configured to guide or focus, to the second refractive member, the light that is reflected or refracted by the first refractive member.

The camera module may further include a dummy member provided between the first refractive member and the second refractive member and configured to guide or focus, to the second refractive member, the light that is reflected or refracted by the first refractive member, wherein the light shielding member is provided on at least a portion of an outer circumferential surface of the dummy member.

According to various embodiments disclosed herein, since the lens assembly and/or the camera module include the folded optics, the telephoto function may be easily implemented in a miniaturized electronic device. For example, since the arrangement direction of the lenses or the forward/backward movement of the lenses for focus adjustment can be implemented substantially parallel to the longitudinal direction or the width direction of the electronic device, a telephoto function can be easily implemented without substantially affecting the thickness of the electronic device. In another embodiment, in the lens assembly and/or camera module, incident light can be refracted at least once (e.g., twice) to reach the image sensor, and the optical image stabilization function can be implemented in the manner of shifting the image sensor in the longitudinal or width direction of the electronic device. For example, it may be easy to secure an installation space of the image sensor and a shift interval for the optical image stabilization function. In another embodiment, as it becomes easy to secure the installation space, it is easy to increase the size and to improve performance of the image sensor, and the performance of the camera can be improved in terms of image quality. In addition, various effects directly or indirectly understood through this document can be provided.

Other aspect(s), advantage(s) and principal configuration(s) of the disclosure will become apparent to those skilled in the art upon reading the following detailed description of various embodiments with reference to the accompanying drawings.

BRIEF DESCRIPTION OF DRAWINGS

The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

FIG. 1 is a block diagram illustrating an electronic device according to various embodiments within a network environment;

FIG. 2 is a perspective view illustrating the front surface of the electronic device according to various embodiments;

FIG. 3 is a perspective view illustrating the rear surface of the electronic device illustrated in FIG. 2;

FIG. 4 is an exploded perspective view illustrating the electronic device illustrated in FIG. 2;

FIG. 5 is a plan view illustrating the rear surface of the electronic device according to various embodiments;

FIG. 6 is a cross-sectional view obtained by cutting a portion of the electronic device along line A-A in FIG. 5;

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FIG. 7 is a configuration view illustrating an optical path of a camera module in the electronic device according to various embodiments;

FIG. 8 is a configuration view illustrating a camera module of the electronic device according to various embodiments; and

FIG. 9 is a configuration view illustrating another example of the camera module of the electronic device according to various embodiments.

Throughout the appended drawings, like reference numerals may be assigned to like parts, components, and/or structures.

DETAILED DESCRIPTION

The following description made with reference to the appended drawings may be provided in order to help comprehensive understanding of the various implementations of the disclosure defined by the claims and their equivalents. A specific embodiment disclosed in the following description includes various specific details to help understanding, but is considered to be one of various embodiments. Accordingly, it will be apparent to those skilled in the art that various changes and modifications of the various implementations described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and configurations may be omitted for clarity and brevity.

The terms and words used in the following description and claims are not limited to a bibliographical meaning, but may be used to clearly and consistently describe the various embodiments disclosed herein. Therefore, it will be obvious to those skilled in the art that the following description of various implementations of the disclosure is provided only for the purpose of explanation, not for the purpose of limiting the disclosure defined as the scope of rights and equivalents thereto.

It should be understood that the singular forms of “a”, “an”, and “the” include plural meanings, unless the context clearly indicate otherwise. Thus, for example, “a component surface” may mean including one or more of component surfaces.

FIG. 1 is a block diagram illustrating an electronic device 101 in a network environment 100 according to various embodiments. Referring to FIG. 1, the electronic device 101 in the network environment 100 may communicate with an electronic device 102 via a first network 198 (e.g., a short-range wireless communication network), or an electronic device 104 or a server 108 via a second network 199 (e.g., a long-range wireless communication network). According to an embodiment, the electronic device 101 may communicate with the electronic device 104 via the server 108. According to an embodiment, the electronic device 101 may include a processor 120, memory 130, an input module 150, a sound output module 155, a display module 160, an audio module 170, a sensor module 176, an interface 177, a connecting terminal 178, a haptic module 179, a camera module 180, a power management module 188, a battery 189, a communication module 190, a subscriber identification module (SIM) 196, or an antenna module 197. In some embodiments, at least one of the components (e.g., the connecting terminal 178) may be omitted from the electronic device 101, or one or more other components may be added in the electronic device 101. In some embodiments, some of the components (e.g., the sensor module 176, the camera

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module 180, or the antenna module 197) may be implemented as a single component (e.g., the display module 160).

The processor 120 may execute, for example, software (e.g., a program 140) to control at least one other component (e.g., a hardware or software component) of the electronic device 101 coupled with the processor 120, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor 120 may store a command or data received from another component (e.g., the sensor module 176 or the communication module 190) in volatile memory 132, process the command or the data stored in the volatile memory 132, and store resulting data in non-volatile memory 134. According to an embodiment, the processor 120 may include a main processor 121 (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor 123 (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor 121. For example, when the electronic device 101 includes the main processor 121 and the auxiliary processor 123, the auxiliary processor 123 may be adapted to consume less power than the main processor 121, or to be specific to a specified function. The auxiliary processor 123 may be implemented as separate from, or as part of the main processor 121.

The auxiliary processor 123 may control, for example, at least some of functions or states related to at least one component (e.g., the display module 160, the sensor module 176, or the communication module 190) among the components of the electronic device 101, instead of the main processor 121 while the main processor 121 is in an inactive (e.g., sleep) state, or together with the main processor 121 while the main processor 121 is in an active (e.g., executing an application) state. According to an embodiment, the auxiliary processor 123 (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module 180 or the communication module 190) functionally related to the auxiliary processor 123. According to an embodiment, the auxiliary processor 123 (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device 101 where the artificial intelligence model is performed or via a separate server (e.g., the server 108). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

The memory 130 may store various data used by at least one component (e.g., the processor 120 or the sensor module 176) of the electronic device 101. The various data may include, for example, software (e.g., the program 140) and

input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or an external electronic device (e.g., an electronic device **102** (e.g., a speaker or a headphone)) directly or wirelessly coupled with the electronic device **101**.

The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, an HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).

The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an

embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify or authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog

beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, an RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and configured to support a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and configured to transmit or receive signals of the designated high-frequency band.

At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional

service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively,” as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

Various embodiments as set forth herein may be implemented as software (e.g., the program) including one or more instructions that are stored in a storage medium (e.g., internal memory or external memory) that is readable by a machine (e.g., the electronic device). For example, a pro-

cessor (e.g., the processor) of the machine (e.g., the electronic device) may invoke at least one of the one or more instructions stored in the storage medium, and execute it. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer’s server, a server of the application store, or a relay server.

According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components or operations may be omitted, or one or more other components or operations may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

In the following detailed description, a longitudinal direction, a width direction, and/or a thickness direction of an electronic device may be referred to, wherein the longitudinal direction may be defined as the “Y-axis direction”, the width direction may be defined as the “X-axis direction”, and/or the thickness direction may be defined as the “Z-axis direction”. In some embodiments, “negative/positive (−/+)” may be referred to together with the Cartesian coordinate system illustrated in the drawings regarding the direction in which a component is oriented. For example, the front surface of an electronic device or a housing may be defined as a “surface oriented in the +Z direction”, and the rear surface may be defined as a “surface oriented in the −Z direction”. In some embodiments, a side surface of an electronic device or a housing may include an area oriented in the +X direction, an area oriented in the +Y direction, an area oriented in the −X direction, and/or an area oriented in

the −Y direction. In another embodiment, the “X-axis direction” may include both the “−X direction” and the “+X direction”. It is noted that these are based on the Cartesian coordinate system described in the drawings for the sake of brevity of description, and the description of these directions or components does not limit the various embodiments disclosed herein.

FIG. 2 is a perspective view illustrating the front surface of the electronic device 200 according to various embodiments. FIG. 3 is a perspective view illustrating the rear surface of the electronic device 200 illustrated in FIG. 2.

Referring to FIGS. 2 and 3, an electronic device 200 according to an embodiment may include a housing 210 including a first surface (or a front surface) 210A, a second surface (or a rear surface) 210B, and a side surface 210C surrounding the space between the first surface 210A and the second surface 210B. In another embodiment, the term “housing” may refer to a structure forming some of the first surface 210A, the second surface 210B, and the side surface 210C of FIG. 2. According to an embodiment, at least a portion of the first surface 210A may be defined by a substantially transparent front plate 202 (e.g., a glass plate or a polymer plate including various coating layers). The second surface 210B may be defined by a substantially opaque rear plate 211. The rear plate 211 may be made of, for example, coated or colored glass, ceramic, a polymer, a metal (e.g., aluminum, stainless steel (STS), or magnesium), or a combination of two or more of these materials. The side surface 210C may be a side structure (or a “side bezel structure”) 218 coupled to the front plate 202 and the rear plate 211 and including a metal and/or a polymer. In some embodiments, the rear plate 211 and the side structure 218 may be integrated with each other, and may include the same material (e.g., a metal material such as aluminum).

In the illustrated embodiment, the front plate 202 may include two first areas 210D, which are bent from the first surface 210A toward the rear plate 211 and extend seamlessly, at the opposite long edges thereof. In FIG. 3, the rear plate 211 may include, at the opposite long edges thereof, two second areas 210E, which are bent from the second surface 210B toward the front plate 202 and extend seamlessly. In some embodiments, the front plate 202 (or the rear plate 211) may include only one of the first areas 210D (or the second areas 210E). In another embodiment, some of the first areas 210D or the second areas 210E may not be included. In the above-described embodiments, when viewed from a side of the electronic device 200, the side structure 218 may have a first thickness (or width) on the side faces, which do not include the first areas 210D or the second areas 210E, and may have a second thickness, which is smaller than the first thickness, on the side faces, which include the first areas 210D or the second areas 210E.

According to an embodiment, the electronic device 200 may include at least one of a display 201, audio modules 203, 207, and 214, sensor modules 204, 216, and 219, camera modules 205, 212, and 213, key input devices 217, light-emitting elements 206, and connector holes 208 and 209. In some embodiments, at least one of the components (e.g., the key input devices 217 or the light-emitting elements 206) may be omitted from the electronic device 200, or other components may be additionally included in the electronic device 200.

The display 201 may be exposed through, for example, a considerable portion of the front plate 202. In some embodiments, at least a portion of the display 201 may be exposed through the front plate 202 defining the first surface 210A and the first areas 210D of the side surfaces 210C. In some

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embodiments, the edges of the display **201** may be configured to be substantially the same as the shape of the periphery of the front plate **202** adjacent thereto. In another embodiment, the distance between the periphery of the display **201** and the periphery of the front plate **202** may be substantially constant in order to enlarge the exposed area of the display **201**.

In another embodiment, recesses or openings may be configured in a portion of the screen display area of the display **201**, and one or more of the audio module **214**, the sensor modules **204**, the camera modules **205**, and the light-emitting elements **206**, which are aligned with the recesses or the openings, may be included. In another embodiment, the rear surface of the screen display area of the display **201** may include at least one of the audio modules **214**, the sensor modules **204**, the camera modules **205**, the fingerprint sensor **216**, and the light-emitting elements **206**. In another embodiment, the display **201** may be coupled to or disposed adjacent to a touch-sensitive circuit, a pressure sensor configured to measure a touch intensity (pressure), and/or a digitizer configured to detect a magnetic-field-type stylus pen. In some embodiments, at least some of the sensor modules **204** and **219** and/or at least some of the key input devices **217** may be disposed in the first areas **210D** and/or the second areas **210E**.

The audio modules **203**, **207**, and **214** may include a microphone hole **203** and speaker holes **207** and **214**. The microphone hole **203** may include a microphone disposed therein to acquire external sound, and in some embodiments, a plurality of microphones may be disposed therein to be able to detect the direction of sound. The speaker holes **207** and **214** may include an external speaker hole **207** and a phone call receiver hole **214**. In some embodiments, the speaker holes **207** and **214** and the microphone hole **203** may be implemented as a single hole, or a speaker (e.g., a piezo speaker) may be included without the speaker holes **207** and **214**.

The sensor modules **204**, **216**, and **219** may generate electrical signals or data values corresponding to the internal operating state or the external environmental state of the electronic device **200**. The sensor modules **204**, **216**, and **219** may include, for example, a first sensor module **204** (e.g., a proximity sensor) and/or a second sensor module (e.g., a fingerprint sensor) disposed on the first surface **210A** of the housing **210**, and/or a third sensor module **219** (e.g., an HRM sensor) and/or a fourth sensor module **216** (e.g., a fingerprint sensor) disposed on the second surface **210B** of the housing **210**. The fingerprint sensor may be disposed not only on the first surface **210A** (e.g., the display **201**) of the housing **210**, but also on the second surface **210B**. The electronic device **200** may further include the sensor module **176** of FIG. 1, for example, at least one of a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

The camera modules **205**, **212**, and **213** may include a first camera device **205** disposed on the first surface **210A** of the electronic device **200**, a second camera device **212** disposed on the second surface **210B** thereof, and/or a flash **213**. The camera devices **205** and **212** may include one or more lenses, an image sensor, and/or an image signal processor. The flash **213** may include, for example, a light-emitting diode or a xenon lamp. In some embodiments, two or more lenses (e.g., an infrared camera lens, a wide-angle lens, and a telephoto lens) and image sensors may be disposed on one surface of the electronic device **200**.

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The key input devices **217** may be disposed on the side surface **210C** of the housing **210**. In another embodiment, the electronic device **200** may not include some or all of the above-mentioned key input devices **217**, and a key input device **217**, which is not included in the electronic device **300**, may be implemented in another form, such as a soft key, on the display **201**. In some embodiments, the key input devices may include a sensor module **216** disposed on the second surface **210B** of the housing **210**.

The light-emitting elements **206** may be disposed, for example, on the first surface **210A** of the housing **210**. The light-emitting elements **206** may provide, for example, information about the state of the electronic device **200** in an optical form. In another embodiment, the light-emitting elements **206** may provide a light source that is interlocked with, for example, the operation of the camera module **205**. The light-emitting elements **206** may include, for example, an LED, an IR LED, and a xenon lamp.

The connector holes **208** and **209** may include a first connector hole **208**, which is configured to accommodate a connector (e.g., a USB connector) for transmitting/receiving power and/or data to/from an external electronic device, and/or a second connector hole **209**, which is configured to accommodate a connector (e.g., an earphone jack) for transmitting/receiving an audio signal to/from an external electronic device.

FIG. 4 is an exploded perspective view illustrating the electronic device **300** illustrated in FIG. 2.

Referring to FIG. 4, the electronic device **300** may include a side structure **310**, a first support member **311** (e.g., a bracket), a front plate **320**, a display **330**, a printed circuit board **340**, a battery **350**, a second support member **360** (e.g., a rear case), an antenna **370**, and a rear plate **380**. In some embodiments, in the electronic device **300**, at least one of the components (e.g., the support member **311** or the second support member **360**) may be omitted, or other components may be additionally included. At least one of the components of the electronic device **300** may be the same as or similar to at least one of the components of the electronic device **200** of FIG. 2 or FIG. 3, and a redundant description thereof will be omitted below.

The first support member **311** may be disposed inside the electronic device **300**, and may be connected to the side structure **310** or may be integrated with the side structure **310**. The first support member **311** may be made of, for example, a metal material and/or a non-metal material (e.g., a polymer). The display **330** may be coupled to one surface of the first support member **311**, and the printed circuit board **340** may be coupled to the other surface of the first support member **411**. On the printed circuit board **340**, a processor, a memory, and/or an interface may be mounted. The processor may include one or more of, for example, a central processing unit, an application processor, a graphics processor, an image signal processor, a sensor hub processor, or a communication processor.

The memory may include, for example, a volatile memory or a nonvolatile memory.

The interface may include, for example, a high-definition multimedia interface (HDMI), a universal serial bus (USB) interface, an SD card interface, and/or an audio interface. The interface may electrically or physically connect, for example, the electronic device **300** to an external electronic device, and may include a USB connector, an SD card/an MMC connector, or an audio connector.

The battery **350** is a device configured to supply power to at least one component of the electronic device **300**, and may include, for example, a non-rechargeable primary battery, a

rechargeable secondary battery, or a fuel cell. At least a portion of the battery 350 may be disposed on substantially the same plane as, for example, the printed circuit board 340. The battery 350 may be integrally disposed inside the electronic device 300, or may be detachably disposed on the electronic device 300.

The antenna 370 may be disposed between the rear plate 380 and the battery 350. The antenna 370 may include, for example, a near field communication (NFC) antenna, a wireless charging antenna, and/or a magnetic secure transmission (MST) antenna. The antenna 370 is configured to, for example, perform short-range communication with an external device or transmitting/receiving power required for charging to/from an external device in a wireless manner. In another embodiment, an antenna structure may be provided by the side structure 310, a portion of the first support member 311, or a combination thereof.

In the following detailed description, reference may be made to the electronic devices 101, 102, 104, 200, and 300 of the embodiments, and the same reference numerals in the drawings are given for components that can be easily understood through the embodiments or omitted, and a detailed description thereof may also be omitted.

FIG. 5 is a plan view illustrating the rear surface of an electronic device 400 (e.g., the electronic device 101, 102, 104, 200, or 300 in FIGS. 1 to 4) according to various embodiments. FIG. 6 is a cross-sectional view obtained by cutting a portion of the electronic device 400 along line A-A' in FIG. 5. FIG. 7 is a configuration view illustrating an optical path of a camera module 500 in the electronic device 400 according to various embodiments.

Referring to FIGS. 5 and 6, the electronic device 400 according to various embodiments may include a camera window 385 disposed on one surface (e.g., the second surface 210B in FIG. 3). In some embodiments, the camera window 385 may be a portion of the rear plate 380. In an embodiment, the camera window 385 may be coupled to the rear plate 380 via a decorative member 389. When viewed from the outside, the decorative member 389 may be exposed in the form of wrapping the periphery of the camera window 385. According to an embodiment, the camera window 385 may include a plurality of transparent areas 387, and the electronic device 400 may receive external light or transmit light to the outside through at least one of the transparent areas 387. For example, the electronic device 400 may include at least one camera module 500 (e.g., the camera module 180, 205, 212, or 213 in FIGS. 1 to 3) disposed to correspond to at least some of the transparent areas 387 and at least one light source (e.g., an infrared light source) disposed to correspond to other ones of the transparent areas 387. In some embodiments, the camera module 500 or the light source may receive external light or emit light to the outside of the electronic device 400 through any one of the transparent areas 387. In an embodiment, the electronic device 400 or the camera module 500 may further include a camera support member 381. The camera support member 381 may allow at least one of the camera module 500 or other camera modules (e.g., a wide-angle camera, an ultra-wide-angle camera, or a macro camera) adjacent to the same to be disposed or fixed inside the rear plate 380 or the camera window 385. In some embodiments, the camera support member 381 may be substantially a portion of the first support member 311 or the second support member 360 of FIG. 4.

According to various embodiments, the electronic device 400 may include at least one of a camera module 500 or a wide-angle camera, an ultra-wide-angle camera, a macro

camera, a telephoto camera, or an infrared photodiode as a light-receiving element, and may include a flash (e.g., the flash 213 in FIG. 3) or an infrared laser diode as a light source or a light-emitting element. In some embodiments, the electronic device 400 may emit an infrared laser toward a subject using an infrared laser diode and an infrared photodiode and receive the infrared laser reflected by a subject to detect a distance or depth to the subject. In another embodiment, the electronic device 400 may photograph a subject by using any one camera or using two or more of the cameras in combination, and may provide illumination toward the subject using a flash, if necessary.

According to various embodiments, among the cameras, a wide-angle camera, an ultra-wide-angle camera, or a close-up camera may have a smaller length in the optical axis (O) direction of the lens(s) when compared to a telephoto camera (e.g., the camera module 500). For example, in a telephoto camera (e.g., the camera module 500) having a relatively large focal length, the total track length of the lens or lenses 423a, 423b, and 423c may be larger than those of other cameras. The total track length may be a distance from the object-side surface of the first object-side lens to the imaging surface of the image sensor. In an embodiment, the wide-angle camera, the ultra-wide-angle camera, or the close-up camera may have substantially little effect on the thickness of the electronic device 400 even if the lens or lenses may be arranged along the thickness of the electronic device 400 (e.g., the thickness measured in the Z-axis direction of FIG. 4 or FIG. 6). For example, a wide-angle camera, an ultra-wide-angle camera, or a close-up camera may be disposed in the electronic device 400 in the state in which a direction in which light is incident from the outside to the electronic device 400 is substantially the same as an optical axis direction of the lens or lenses. In another embodiment, when compared to a wide-angle camera, an ultra-wide-angle camera, or a close-up camera, the camera module 500 (e.g., a telephoto camera) has a smaller angle of view, but may be useful for photographing a subject from a greater distance, and may include more lenses 421a, 421b, 423a, 423b, and 423c. For example, when the lens or lenses 423a, 423b, and 423c of the camera module 500 may be arranged in the thickness direction of the electronic device 400 (e.g., the Z-axis direction), the thickness of the electronic device 400 may increase or, the camera module 500 may substantially protrude to the outside of the electronic device 400. In various embodiments, the camera module 500 may include at least one refractive member 413 or 415 that reflects or refracts incident light IL in different directions. In implementing a telephoto function, the lenses 423a, 423b, and 423c may be arranged to move forward or backward in the incident direction of light or the traveling direction of reflected or refracted light, thereby preventing or reducing the increase of the thickness of the electronic device 400.

Referring to FIGS. 6 and 7, the folded camera (e.g., the camera module 500) may include a first refractive member 413, a second refractive member 415, an image sensor 411, and/or at least one optical lens (e.g., the second lens group 423 including the second lenses 423a, 423b, and 423c or the dummy member 423d). In an embodiment, the at least one optical member may guide or focus, to the second refractive member 415, light RL1 reflected or refracted by the first refractive member 413, and may block the light RL1 reflected or refracted the first refractive member 413 from being directly incident on the image sensor 411. In general, the term "optical member" may be a component configured to transmit, refract, scatter, absorb, or reflect light. In describing various embodiments disclosed herein, the term

optical member may refer to the second lenses **423a**, **423b**, and **423c** or the dummy member **423d** disposed between the first refractive member **413** and the second refractive member **415**. However, it is noted that this term or expression does not limit the various embodiments. For example, the first refractive member **413**, the second refractive member **415**, and/or the first lenses **421a** and **421b** (or the first lens group **421**) may be types of the optical member. In the following detailed description, if necessary, the optical member disposed between the first refractive member **413** and the second refractive member **415** may be separately described as the second lens or lenses and/or the dummy member, and the second lenses **423a**, **423b**, **423c** or the dummy member **423d** may be described as a second lens group **423** in combination.

According to various embodiments, the first refractive member **413** may include, for example, a prism or a mirror, and may reflect or refract light **IL** that is incident in the first direction **D1**, in the second direction **D2** intersecting the first direction **D1**. The first direction **D1** may be, for example, the direction in which light **IL** is incident on the electronic device **400** or the camera module **500** from the outside through at least one of the transparent areas **387** of FIG. **5** when photographing a subject. In some embodiments, the first direction **D1** may be a photographing direction, a subject direction, an orientation direction of the camera module **500**, or a direction parallel thereto. In another embodiment, the first direction **D1** may be parallel to the thickness direction of the electronic device **400** or the Z-axis direction. According to an embodiment, when the second direction **D2** is perpendicular to the first direction **D1**, it may be more easy to design the camera module **500** or an optical path. For example, the second direction **D2** may be parallel to the longitudinal direction of the electronic device **400** (e.g., the Y-axis direction) or the width direction of the electronic device **400** (e.g., the X-axis direction). However, embodiments are not limited thereto. For example, the second direction **D2** may be designed in an inclined direction rather than a direction perpendicular to the first direction **D1** depending on the arrangement and specifications of the camera module **500** or the first refractive member **413** in the electronic device **400**, and may be a direction inclined with respect to the X-axis or the Y-axis while being parallel to the X-Y plane.

According to various embodiments, the second refractive member **415** may include, for example, a prism or a mirror, and may reflect or refract light **RL1** that is reflected or refracted by the first refractive member **413** and is incident in the second direction **D2**, in the third direction **D3** intersecting the second direction **D2**. In an embodiment, the third direction **D3** may be substantially perpendicular to the second direction **D2**. For example, the third direction **D3** may be a direction parallel to the Z-axis direction. However, embodiments are not limited thereto, and the third direction **D3** may be a direction inclined with respect to the second direction **D2** or the X-Y plane depending on the arrangement and specifications of the camera module **500** or the second refractive member **415** in the electronic device **400**. In another embodiment, the third direction **D3** may be parallel to the first direction **D1**.

According to various embodiments, the image sensor **411** may be configured to detect the light **RL2** that is reflected or refracted by the second refractive member **415** and is incident along the third direction **D3**. For example, the light **IL** incident from the outside may be detected by the image sensor **411** via the first refractive member **413** and the second refractive member **415**, and the electronic device **400**

or the camera module **500** may acquire a subject image based on a signal or information detected through the image sensor **411**. In an embodiment, the image sensor **411** may be disposed substantially parallel to the X-Y plane. For example, when the camera module **500** has an optical image stabilization function of a structure that shifts the image sensor **411**, the image sensor **411** may move horizontally in a plane perpendicular to the first direction **D1** or the third direction **D3**.

According to various embodiments, in performing the optical image stabilization function, the image sensor **411** may be shifted in the longitudinal direction of the electronic device **400** (e.g., Y-axis direction) or the width direction of the electronic device **400** (e.g., X-axis direction). For example, since the image sensor **411** is disposed on a plane perpendicular to the first direction **D1** or the third direction **D3**, it may be easier to increase the size of the image sensor in an electronic device having a small thickness (e.g., a thickness within about 10 mm) and/or to secure a space for the optical image stabilization operation. In an embodiment, when the camera module **500** is used as a telephoto camera, the quality of a captured image may be further enhanced by being provided with the optical image stabilization function. In another embodiment, when the image sensor **411** is enlarged, the performance of the camera module **500** may be further enhanced.

According to various embodiments, the camera module **500** may further include a lens group (e.g., the first lens group **421** including one or more first lenses **421a** and **421b**) that guides or focuses, to the first refractive member **413**, the light **IL** that is incident in the first direction **D1**, wherein the at least one optical member may be disposed between the first refractive member **413** and the second refractive member **415** to provide a second lens group **423**. In an embodiment, the first lens group **421** or the first lens (e.g., the first lens **421a**) disposed on the object side in the camera module **500** may have a positive refractive power. For example, by configuring the first lens **421a** to focus or align the light **IL** that is incident from the outside, to the first refractive member **413**, the optics from the first lens **421a** to the image sensor **411** may be miniaturized. According to an embodiment, the first lens group **421** may further include an additional first lens or lenses **421b** in order to focus or align light incident from the outside.

According to various embodiments, at least one optical member (e.g., the second lens group **423**) may include a dummy member **423d** and a light shielding member **425**. The dummy member **423d** may have, for example, a cylinder shape disposed inside the camera module **500** or the electronic device **400** and extending along the second direction **D2**, and may transmit the light **RL2** traveling along the second direction **D2**. In some embodiments, the dummy member **423d** may be one lens having a positive or negative refractive power. In another embodiment, the dummy member **423d** may be a component integrated with any one of the second lenses **423a**, **423b**, and **423c** or the second refractive member **415**.

According to various embodiments, the light shielding member **425** may be provided or disposed on at least a portion of the outer circumferential surface of the dummy member **423d**, and may absorb, scatter, or reflect light. The light shielding member **425** may be provided by performing, for example, etching or black lacquer processing, and/or printing or depositing a reflective layer on at least a portion of an outer peripheral surface of the dummy member **423d**. In an embodiment, a part of the light reflected or refracted by the first refractive member **413** (e.g., the light indicated

by “GRL” in FIG. 8) may be absorbed, scattered, or reflected by the light shielding member 425, and the light shielding member 425 may substantially block the light that is reflected or refracted by the first refractive member 413 (the light indicated by “RL1” in FIG. 7 or “GRL” in FIG. 8) from being directly incident on the image sensor 411.

According to various embodiments, light may be incident on the camera module in a direction other than the photographing direction or the orientation direction (GIL, see FIG. 8) or may travel through a path other than a designed optical path (e.g., the paths indicated by “RL1” and/or “RL2” in FIG. 7) in the process of reflection or refraction within the camera module 500, thereby being detected by the image sensor. Light incident on the image sensor through an unintended path or an image generated by the light may be referred to as “ghost light” or “ghost image”. The quality of an image acquired through the camera module 500 may be deteriorated due to the ghost image. According to an embodiment, the light shielding member 425 is configured to absorb, scatter, or reflect ghost light (e.g., the light GRL reflected or refracted by the first refractive member 413), thereby blocking the ghost light from being directly incident on the image sensor 411. For example, light sequentially passing through the first direction D1, the second direction D2, and/or the third direction D3 in the camera module 500 (e.g., the light following the paths indicated by “IL”, “RL1”, and “RL2” in FIG. 7) may be incident on the image sensor 411, and the light traveling another path may be substantially blocked from being incident on the image sensor 411.

According to various embodiments, the at least one optical member may include at least one second lenses 423a, 423b, or 423c moving forward and backward along the second direction D2 between the first refractive member 413 and the second refractive member 415. For example, the electronic device 400 or the camera module 500 may adjust the focal length by moving the at least one second lens 423a, 423b, or 423c forward or backward in the second direction D2. A miniaturized electronic device such as a smart phone may have a thickness of about 10 mm, and in this case, a range in which the lens can move forward and backward in the thickness direction may be limited.

According to various embodiments, the second direction D2 can be substantially parallel to the longitudinal direction (e.g., the Y-axis direction of FIG. 4), the width direction (e.g., the X-axis direction of FIG. 4), and/or the X-Y plane, and compared to a general wide-angle camera that moves forward and backward in the Z-axis direction for focus adjustment, the range in which the at least one second lens 423a, 423b, or 423c may move forward and backward may be large. For example, since at least one second lens 423a, 423b, or 423c moves forward and backward along the second direction D2, the telephoto performance may be improved in the camera module 500, and thus the degree of design in securing a space for forward and backward movement for focus adjustment can be improved.

According to various embodiments, the effective focal length of the lens assembly or camera module 500 may be about 1.8 times or more of the diagonal length of the imaging surface or active area of the image sensor 411 (hereinafter, the “size of the image sensor”). For example, when mounted in a miniaturized electronic device having a length or width of about 180 mm or less and a thickness of about 10 mm or less (e.g., the electronic device 300 or 400 in FIG. 4 or 6), the ratio of the size and the effective focal length of the image sensor 411 described above may be suitable for enlarging the image sensor 411 while implementing a telephoto function. In an embodiment, when the

camera module 500 is manufactured to have an effective focal length of about 11.7 mm and the image sensor 411 is manufactured to have a size of about 5.5 mm, the performance of the image sensor 411 can be enhanced while implementing a telephoto function. Here, the “size” of the image sensor 411 may be a diagonal length of an effective pixel area (an active area). In another embodiment, in a miniaturized electronic device, considering the mounting space of an electronic component such as a battery (e.g., the battery 350 in FIG. 4) or a circuit board (e.g., the printed circuit board 340 in FIG. 4), the effective focal length of the camera module 500 may be approximately 2.5 times or less of the size of the image sensor 411.

According to various embodiments, the electronic device 400 and/or the camera module 500 may further include an infrared cut-off filter 419. In an embodiment, the infrared cut-off filter 419 may block light of an infrared or near-infrared wavelength band from being incident on the image sensor 411 and may be disposed at any position in the optical path between the first lens 421a and the image sensor 411. In some embodiments, by disposing the infrared cut-off filter 419 at a position close to the image sensor 411 (e.g., between the image sensor 411 and the second refractive member 415), it is possible to inhibit or prevent the infrared cut-filter 419 from being visually exposed to the outside. In an embodiment, the first refractive member 413, the second refractive member 415, and/or the at least one optical member (e.g., the second lens group 423) may include an infrared cut-off coating layer, and in this case, the infrared cut-off filter 419 may be omitted. In some embodiments, the infrared cut-off coating layer may be provided on at least one of the image sensor-side surface and the object-side surface of the dummy member 423d or the second refractive member 415. Accordingly, the image sensor 411 may detect light that substantially passes through the infrared cut-off filter 419 (or the infrared cut-off coating layer).

According to various embodiments, the electronic device 400 and/or the camera module 500 may not include an optical member, for example, the dummy member 423d. In this case, a light shielding member 425 may be disposed on at least a portion of the infrared cut-off filter 419. Various arrangements of the light shielding member 425 will be further described with reference to FIGS. 8 and 9.

FIG. 8 is a cross-sectional view illustrating a camera module (e.g., the camera module 500 in FIG. 6) of an electronic device (e.g., the electronic device 300 or 400 of FIG. 4 or FIG. 6) according to various embodiments. FIG. 9 is a configuration view illustrating another example of the camera module 500 in the electronic device 400 according to various embodiments.

Referring to FIGS. 8 and 9, the light shielding member 425 may be disposed on the infrared cut-off filter 419 or the dummy member 423d. As described above, the dummy member 423d may be disposed between the first refractive member 413 and the second refractive member 415, and the infrared cut-off filter 419 may be disposed between the second refractive member 415 and the image sensor 411. According to an embodiment, light focused or aligned by the first lens 421a (or the first lens group 421) (e.g., light indicated by “IL” in FIG. 7) may be incident on the first refractive member 413, and the light refracted or reflected by the first refractive member 413 (e.g., the light indicated by “RL1” in FIG. 7) may be generally guided to the second refractive member 415. However, a part of the incident light (e.g., the light indicated by “GIL” in FIGS. 8 and 9) may be refracted or reflected by the first refractive member 413 to travel toward the image sensor 411. In an embodiment, the

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light shielding member **425** may be disposed on the infrared cut-off filter **419** or the dummy member **423d** and may substantially block the light refracted or reflected by the first refractive member **413** (e.g., the light indicated by “GIL” in FIGS. **8** and **9**) from being directly incident on the image sensor. For example, the light shielding member **425** may absorb, scatter, or reflect light that is refracted or reflected by the first refractive member **413** and travels toward the image sensor **411**, thereby directly blocking the light from being directly incident on the image sensor **411** and suppressing the occurrence of a ghost image.

According to various embodiments, when the camera module **500** includes a barrel structure **429** configured to dispose therein the second lens or lenses **423a**, **423b**, and **423c(s)**, the dummy member **423d** may be omitted and the barrel structure **429** itself may provide the function of the light shielding member **425** or the light shielding member **425** may be disposed in the barrel structure **429**. For example, the dummy member **423d** may be omitted, and the infrared cut-off filter **419** or the barrel structure **429** may provide an environment in which the light shielding member **425** may be disposed. In another embodiment, when any one of the second lenses **423a**, **423b**, and **423c** is disposed at an appropriate position and the outer circumferential surface thereof has a sufficient length in the second direction **D2**, the light shielding member **425** may be provided on any one of the second lens **423a**, **423b**, and **423c**. In another embodiment, although a name different from the second lenses **423a**, **423b**, and **423c** is given, the dummy member **423d** may function as an additional second lens when having a positive or negative refractive power. When it is described that the dummy member **423d** “functions as a lens”, it may be that the dummy member **423d** aligns, focuses, or guides transmitted light. In another embodiment, in a structure in which the infrared cut-off filter **419** is disposed considerably close to the image sensor **411**, when the light shielding member **425** is disposed on the infrared cut-off filter **419** in the third direction (e.g., the third direction **D3** in FIG. **6**), the light shielding member **425** may interfere with the optical path leading to the image sensor **411**. For example, in disposing the light shielding member **425** in the infrared cut-off filter **419**, the shape or position of the light shielding member **425** may be determined in consideration of the optical path between the second refractive member **415** and the image sensor **411**.

In the camera module **500** according to various embodiments, light incident from the outside may be reflected or refracted at least once to reach the image sensor **411**. For example, since it is possible to dispose the arrangement direction of lenses or the movement direction of a lens (e.g., the second lens(es) **423a**, **423b**, and **423c** in FIG. **6** or **7**) for adjusting the focal length parallel to the longitudinal direction of the electronic device **400** (e.g., the Y-axis direction in FIG. **4**) or the width direction of the electronic device **400** (e.g., the X-axis direction in FIGS. **4** and **6**), it may be easier to provide a telephoto function when the camera module **500** is mounted in a miniaturized electronic device. In another embodiment, with the optical path having a structure in which light is reflected or refracted, it is possible to provide an environment that enables the image sensor **411** to be disposed on a plane substantially parallel to the longitudinal direction and/or the width direction of the electronic device **400**, and with the structure in which the image sensor **411** is shifted, it is possible to provide an optical image stabilization structure. For example, the camera module **500** according to various embodiments disclosed herein is configured to provide a telephoto function and a handshake correction

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function when mounted on a miniaturized electronic device (e.g., the electronic devices **300** and **400** of FIG. **4** or **6**). In another embodiment, when the image sensor **411** is disposed in a plane substantially parallel to the longitudinal direction and/or the width direction of the electronic device **400**, it may be easier to increase the size of the image sensor **411** when compared to a structure in which the image sensor is disposed in a plane parallel to the thickness direction of the electronic device. For example, since the image sensor **411** is disposed in a plane parallel to the longitudinal direction and/or the width direction of the electronic device, it may be easier to mount the high-performance camera module **500** in a miniaturized electronic device.

As described above, according to various embodiments, a lens assembly and/or a camera module (e.g., the camera module **500** in FIGS. **6** to **9**) may include a first refractive member (e.g., the first refractive member **413** of FIGS. **6** to **9**) configured to reflect or refract light (e.g., the light indicated by “IL” in FIG. **6**) that is incident in a first direction (e.g., the first direction **D1** in FIG. **6**), in a second direction (e.g., the second direction **D2** in FIG. **6**) intersecting the first direction, a second refractive member (e.g., the second refractive member **415** in FIGS. **6** to **9**) configured to reflect or refract the light (e.g., the light indicated by “RL1” in FIG. **6**) that is reflected or refracted by the first refractive member and is incident along the second direction, in a third direction (e.g., the third direction **D3** in FIG. **6**) intersecting the second direction, an image sensor (e.g., the image sensor **411** in FIGS. **6** to **9**) configured to detect the light that is reflected or refracted by the second refractive member and is incident along the third direction (e.g., the light indicated by “RL2” in FIG. **6**), and at least one optical member (e.g., the second lens or lenses **423a**, **423b**, and **423c** or dummy member **423d** in FIGS. **6** to **9**) disposed in an area or space between the first refractive member and the second refractive member. The at least one optical member may be configured to block the light that is reflected or refracted by the first refractive member from being incident on the image sensor, and to guide light incident on the second refractive member along the second direction.

According to various embodiments, the at least one optical member may include a dummy member (e.g., the dummy member **423d** in FIG. **6** or FIG. **9**) extending along the second direction and transmitting light traveling in the second direction, and a light shielding member (e.g., the light shielding member **425** in FIG. **6**, FIG. **8**, and/or FIG. **9**) provided on at least a portion of an outer circumferential surface of the dummy member, and the light shielding member may be configured to absorb, scatter or reflect the light (e.g., the light indicated by “GRL” in FIG. **8** or FIG. **9**) that is reflected or refracted by the first refractive member to block the light from being incident on the image sensor.

According to various embodiments, the above-described camera module may further include a lens group (e.g., the first lens **421a** or the first lens group **421** in FIGS. **6** to **9**) having a positive refractive power, and the lens group may be configured to guide or focus, to the first refractive member, the light that is incident in the first direction.

According to various embodiments, at least one optical member may include at least one lens (e.g., at least one of the second lenses **423a**, **423b**, and **423c** in FIGS. **6** to **9**) configured to move forward and backward between the first refractive member and the second refractive member along the second direction.

According to various embodiments, at least one optical member may include a light shielding member provided on at least a portion of an outer circumferential surface of the

at least one lens, and the light shielding member may be configured to absorb, scatter, or reflect the light that is reflected or refracted by the first refractive member to block the light from being incident on the image sensor.

According to various embodiments, the above-described camera module may include an infrared cut-off filter (e.g., the infrared cut-off filter **419** in FIGS. **6** to **9**) disposed between the second refractive member and the image sensor, and the image sensor may be configured to detect light that passes through the infrared cut-off filter.

According to various embodiments, at least one of the first refractive member, the second refractive member, or the at least one optical member may include an infrared cut-off coating layer, and the image sensor may be configured to detect light that passes through the infrared cut-off coating layer.

According to various embodiments, the first refractive member may be configured to reflect or refract light in a direction perpendicular to the first direction, and the second refractive member is configured to reflect or refract light in a direction perpendicular to the second direction.

According to various embodiments, the image sensor may be configured to horizontally move in a plane perpendicular to the first direction or a plane perpendicular to the third direction (e.g., the X-Y plane in the coordinate system of FIG. **4** or a plane parallel to the same).

According to various embodiments, in the above-described camera module, the effective focal length may be set to be 1.8 times or more and 2.5 times or less of the diagonal length of the imaging plane of the image sensor.

According to various embodiments disclosed herein, a camera module (e.g., the camera module in FIGS. **6** to **9**) may include a first refractive member (e.g., the first refractive member **413** of FIGS. **6** to **9**) configured to reflect or refract light (e.g., the light indicated by “IL” in FIG. **6**) that is incident in a first direction (e.g., the first direction **D1** in FIG. **6**), in a second direction (e.g., the second direction **D2** in FIG. **6**) intersecting the first direction, a second refractive member (e.g., the second refractive member **415** in FIGS. **6** to **9**) configured to reflect or refract the light (e.g., the light indicated by “RL1” in FIG. **6**) that is reflected or refracted by the first refractive member and is incident along the second direction, in a third direction (e.g., the third direction **D3** in FIG. **6**) intersecting the second direction, an image sensor (e.g., the image sensor **411** in FIGS. **6** to **9**) configured to detect the light that is reflected or refracted by the second refractive member and is incident along the third direction (e.g., the light indicated by “RL2” in FIG. **6**), and a light shielding member (e.g., the light shielding member **425** in FIG. **6**, FIG. **8**, and/or FIG. **9**) disposed in an area or a space between the first refractive member and the second refractive member. The light shielding member may be configured to at least partially absorb, scatter, or reflect the light that is reflected or refracted by the first refractive member and travels toward the image sensor.

According to various embodiments, the above-described camera module may further include a lens group (e.g., the first lens **421a** or the first lens group **421** in FIGS. **6** to **9**) having a positive refractive power, and the lens group may be configured to guide or focus, to the first refractive member, the light that is incident in the first direction.

According to various embodiments, the above-described camera module may further include at least one lens (e.g., at least one of the second lenses **423a**, **423b**, and **423c** in FIGS. **6** to **9**) configured to move forward and backward between the first refractive member and the second refractive member along the second direction, and at least one lens may be

configured to guide or focus, to the second refractive member, the light that is reflected or refracted by the first refractive member.

According to various embodiments, the image sensor may be configured to horizontally move in a plane perpendicular to the first direction or a plane perpendicular to the third direction (e.g., the X-Y plane in the coordinate system of FIG. **4** or a plane parallel to the same).

According to various embodiments, the above-described camera module may include an infrared cut-off filter (e.g., the infrared cut-off filter **419** in FIGS. **6** to **9**) disposed between the second refractive member and the image sensor, and the light shielding member is disposed on at least a portion of the infrared cut-off filter, and the image sensor may be configured to detect light that passes through the infrared cut-off filter.

According to various embodiments, the above-described camera module may include a dummy member (e.g., the dummy member **423d** in FIG. **6** or FIG. **9**) disposed between the first refractive member and the second refractive member and configured to guide or focus, to the second refractive member, the light that is reflected or refracted by the first refractive member, and the light shielding member may be provided on at least a portion of an outer circumferential surface of the dummy member.

According to various embodiments, the first refractive member may be configured to reflect or refract light in a direction perpendicular to the first direction, and the second refractive member is configured to reflect or refract light in a direction perpendicular to the second direction.

According to various embodiments, the above-described camera module may include a lens group having a positive refractive power and configured to guide or focus, to the first refractive member, the light incident in the first direction, and at least one lens disposed between the first refractive member and the second refractive member and configured to move forward and backward along the second direction. The at least one lens may be configured to guide or focus, to the second refractive member, the light that is reflected or refracted by the first refractive member.

According to various embodiments, the above-described camera module may include a dummy member disposed between the first refractive member and the second refractive member and configured to guide or focus, to the second refractive member, the light that is reflected or refracted by the first refractive member, and the light shielding member may be provided on at least a portion of an outer circumferential surface of the dummy member.

According to various embodiments, in the above-described camera module, the effective focal length may be set to be 1.8 times or more and 2.5 times or less of the diagonal length of the imaging plane of the image sensor.

According to various embodiments, since the lens assembly and/or the camera module include the folded optics, the telephoto function may be easily implemented in a miniaturized electronic device. For example, since the arrangement direction of the lenses or the forward/backward movement of the lenses for focus adjustment can be implemented substantially parallel to the longitudinal direction or the width direction of the electronic device, a telephoto function can be easily implemented without substantially affecting the thickness of the electronic device. In another embodiment, in the lens assembly and/or camera module, incident light can be refracted at least once (e.g., twice) to reach the image sensor, and the optical image stabilization function can be implemented in the manner of shifting the image sensor in the longitudinal or width direction of the electronic

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device. For example, it may be easy to secure an installation space of the image sensor and a shift interval for the optical image stabilization function. In another embodiment, as it becomes easier to secure the installation space, it is easier to increase the size and to improve performance of the image sensor, and the performance of the camera can be improved in terms of image quality. In addition, various effects directly or indirectly understood through this document can be provided.

In the foregoing detailed description, embodiments have been described. However, it will be evident to a person ordinarily skilled in the art that various modifications can be made without departing from the scope of the disclosure.

What is claimed is:

1. A camera module comprising:
 - a first refractive member configured to reflect or refract light that is incident on the first refractive member in a first direction, in a second direction intersecting the first direction;
 - a second refractive member configured to reflect or refract the light that is reflected or refracted by the first refractive member and is incident on the second refractive member in the second direction, in a third direction intersecting the second direction;
 - an image sensor configured to detect the light that is reflected or refracted by the second refractive member and is incident on the image sensor in the third direction; and
 - at least one optical member provided between the first refractive member and the second refractive member, wherein the at least one optical member comprises a dummy member extending in the second direction and configured to transmit light traveling in the second direction, and a first light shielding member provided on at least a portion of an outer circumferential surface of the dummy member, wherein the dummy member is configured to guide the light that is reflected or refracted by the first refractive member to be incident on the second refractive member in the second direction, wherein the first light shielding member and the outer circumferential surface of the dummy member face the third direction, and wherein the first light shielding member is configured to absorb, scatter or reflect the light that is reflected or refracted by the first refractive member to block a portion of the light from being directly incident on the image sensor.
2. The camera module of claim 1, further comprising a lens group having a positive refractive power, wherein the lens group is configured to guide or focus, to the first refractive member, the light that is incident in the first direction.
3. The camera module of claim 1, wherein the at least one optical member further comprises at least one lens configured to move forward and backward between the first refractive member and the dummy member along the second direction.
4. The camera module of claim 3, wherein the at least one optical member further comprises a second light shielding member provided on at least a portion of an outer circumferential surface of the at least one lens, and wherein the second light shielding member is configured to absorb, scatter, or reflect the light that is reflected or refracted by the first refractive member to block the light from being directly incident on the image sensor.

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5. The camera module of claim 1, further comprising an infrared cut-off filter provided between the second refractive member and the image sensor,

wherein the image sensor is configured to detect light that passes through the infrared cut-off filter.

6. The camera module of claim 1, wherein at least one of the first refractive member, the second refractive member, or the at least one optical member comprises an infrared cut-off coating layer, and

wherein the image sensor is configured to detect light that passes through the infrared cut-off coating layer.

7. The camera module of claim 1, wherein the second direction is perpendicular to the first direction, and the third direction is perpendicular to the second direction.

8. The camera module of claim 1, wherein the image sensor is configured to horizontally move in a plane perpendicular to the first direction or a plane perpendicular to the third direction.

9. The camera module of claim 1, wherein an effective focal length is 1.8 times or more and 2.5 times or less of a diagonal length of an imaging plane of the image sensor.

10. The camera module of claim 1, wherein the first light shielding member is configured to at least partially absorb, scatter, or reflect the light that is reflected or refracted by the first refractive member and travels toward the image sensor other than the second direction.

11. The camera module of claim 10, further comprising at least one lens configured to move forward and backward along the second direction between the first refractive member and the dummy member,

wherein the at least one lens and the dummy member is configured to guide or focus, to the second refractive member, the light that is reflected or refracted by the first refractive member.

12. The camera module of claim 10, further comprising: an infrared cut-off filter provided between the second refractive member and the image sensor, and

a third light shielding member is provided on at least a portion of an outer circumferential surface of the infrared cut-off filter,

wherein the image sensor is configured to detect light that passes through the infrared cut-off filter.

13. The camera module of claim 10, further comprising: a lens group having a positive refractive power and configured to guide or focus, to the first refractive member, the light incident on the first refractive member in the first direction; and

at least one lens provided between the first refractive member and the dummy member, the at least one lens being configured to move forward and backward along the second direction,

wherein the at least one lens and the dummy member are configured to guide or focus, to the second refractive member, the light that is reflected or refracted by the first refractive member.

14. An electronic device comprising:

a housing; and

a camera module provided on the housing, the camera module comprising:

a first refractive member configured to reflect or refract light that is incident on the first refractive member in a first direction, in a second direction intersecting the first direction;

a second refractive member configured to reflect or refract the light that is reflected or refracted by the first refractive member and is incident on the second refrac-

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tive member in the second direction, in a third direction intersecting the second direction;

an image sensor configured to detect the light that is reflected or refracted by the second refractive member and is incident on the image sensor in the third direction; and

at least one optical member provided between the first refractive member and the second refractive member, wherein the at least one optical member comprises a dummy member extending in the second direction and configured to transmit light traveling in the second direction, and a first light shielding member provided on at least a portion of an outer circumferential surface of the dummy member,

wherein the dummy member is configured to guide the light that is reflected or refracted by the first refractive member to be incident on the second refractive member in the second direction,

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wherein the first light shielding member and the outer circumferential surface of the dummy member face the third direction, and

wherein the first light shielding member is configured to absorb, scatter or reflect the light that is reflected or refracted by the first refractive member to block a portion of the light from being directly incident on the image sensor.

15. The electronic device of claim **14**, further comprising a lens group having a positive refractive power, wherein the lens group is configured to guide or focus, to the first refractive member, the light that is incident in the first direction.

16. The electronic device of claim **14**, wherein the at least one optical member comprises at least one lens configured to move forward and backward between the first refractive member and the second refractive member along the second direction.

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